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ABSTRACT

As data availability improves, the gravity model of international trade is used to study the flow of trade over increasingly long periods of time. This puts pressure on the standard assumption that constant bilateral fixed effects are sufficient to control for unobserved heterogeneity. When looking across decades rather than years, changes in the economic landscape will affect both explanatory and dependent variables, thus biasing the results. The paper begins by providing evidence of time variation in the unobserved heterogeneity using a novel econometric test. It then presents a new way of estimating the gravity model that can identify and control for unobserved heterogeneity as it varies over country pairs and time. This approach substantially enhances the precision of the estimated parameters, enabling a detailed analysis of the impact of different types of economic integration agreements on trade on an annual basis. This heightened precision also demonstrates the diminishing importance of distance and allows our model to accurately depict the evolution of trade over time. The pattern in the unobserved heterogeneity closely resembles the ascent and the deceleration of globalization. The paper closes with some preliminary analysis into the factors that underlie the unobserved heterogeneity.

1. Introduction

In the fifty years between the end of the Second World War and the beginning of the new millennium, the world has witnessed enormous growth in global trade, which has easily outpaced the increase in global productivity. This phenomenon, often referred to as globalization, has most likely been driven by a steady decline in the cost of trade. Although trade continued to grow into the early part of the new millennium, globalization slowed considerably following the onset of the Great Recession (2008) and the Global Trade Collapse (2009).

This rise and subsequent slowdown of globalization has been well documented in the international trade literature.¹ Still, our standard empirical models do not allow us to fully account for the historical evolution of the globalization process. While the gravity model of international trade explains much of the cross-sectional variation in trade flows, it has severe difficulties capturing how they change over time (Lai and Trefler, 2002). The main reason for this is that most variables in the trade-cost equation are time-invariant and therefore cannot account for trade growth. In addition, those variables that vary over time, such as tariffs and trade

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¹ Krugman (1995) and Feenstra (1998) were two of the earlier researchers who detailed the possible sources of the growth in trade; these sources included falling transportation and information costs, falling tariffs, and the growth of offshoring. For a more recent treatment, see Redding and Turner (2015) and Ganapati and Wong (2023).

agreements, have not been able to explain much of the growth in trade or its heterogeneity across bilateral pairs. Even when we do allow standard gravity variables, like distance, to interact with time, the model still cannot account for much of trade growth. This implies that bilateral trade flows may be substantially influenced by unobserved factors that vary over time, which, by definition, cannot be captured by the constant bilateral fixed effects that most empirical gravity models include.

Without accounting for the globalization process, coefficient estimates of these policy variables might be biased if trade policies respond to trade growth or globalization. We propose a novel estimation strategy that can control for an unobserved globalization process in which each bilateral pair of trade partners can have a heterogeneous response. To identify this unobserved factor, we employ a Kalman filter. Most often used in macroeconomics, the Kalman filter was popularized by [Stock and Watson \(2001\)](#). They employed this technique to identify common, unobserved components from a set of macroeconomic variables. In our setting, we employ the Kalman filter to allow us to model a common trend in the international trade system that has an asymmetric impact on each bilateral pair without imposing strong restrictive assumptions. This way, we let the empirical data speak for themselves. Most notably, while our model chooses to remain agnostic about the behavior of this trend over time, we find that the estimated parameter follows a relatively smooth pattern that looks similar to the globalization process described at the outset of this paper.

The paper makes three contributions to the literature. First, we provide a more flexible approach to deal with unobserved heterogeneity. The standard assumption in the empirical gravity literature is that bilateral effects alone can account for the potential endogeneity of economic integration agreements (EIAs). Initially explored in [Baier and Bergstrand \(2007\)](#), nearly every panel gravity equation that estimates the impact of EIAs includes bilateral fixed effects that are presumed to be correlated with the decision to enter into the agreement. As long as the unobserved factors are constant or slow-moving over time, the bilateral fixed effect can account for the potential endogeneity and produce unbiased coefficient estimates on the EIAs. When much longer time horizons come into play, as they do in the most recent panel gravity models, this changes. On timescales that span half a century, the changing economic landscape will inevitably impact the decision of member states to enter into EIAs. This renders time-invariant bilateral fixed effects insufficient and may lead to biased parameter estimates. Using a novel econometric test, we first provide evidence for the presence of these unobserved time-varying coefficients associated with bilateral pair heterogeneity. Building on the work of [Baier et al. \(2014\)](#) and [Bergstrand et al. \(2015\)](#), we then model and estimate a gravity model that includes heterogeneous responses to the globalization process.² Thanks to the use of a Kalman filter, we no longer have to assume this process to be linear over time.

Second, where most of the research in the field has used panel data with intervals of three to five years ([Cheng and Wall, 2005](#)), this paper traces the impact of different types of EIAs on a yearly basis. While the use of annual data is not new, instability in the estimates required additional simplifications. [Baier et al. \(2014\)](#), e.g., obtained noisy estimates of the impact of EIAs over time. They addressed this problem by assuming a linear trend before and the agreements' enactment dates. More recently, [Egger et al. \(2022\)](#) used annual data to show the potential bias resulting from interval estimates. However, their analysis combined free trade agreements, customs unions, common markets, and economic unions into a single dummy variable. By allowing time-varying unobserved coefficients to interact with bilateral specific effects, our estimator resolves a source of endogeneity, resulting in more precise coefficient estimates of EIAs. The increased precision allows us to measure the differential effects of each type of agreement and trace out their impact for up to 25 years without having to impose any additional assumptions.

Third, our estimation model contributes to solving the so-called 'distance puzzle' by allowing for greater heterogeneity in the response to globalization. The distance puzzle is the paradox between conventional wisdom, which would expect the distance elasticity to fall as global trade increases, and the empirical data, which show that the (absolute value of the) distance elasticity has instead increased ([Disdier and Head, 2008](#)). [Yotov \(2012\)](#) was the first to provide a solution to this distance puzzle by introducing heterogeneity into the evolution of trade costs in his model. He showed that the distance puzzle can be resolved by including domestic trade flows and allowing these and international trade flows to evolve at different rates. The critical insight in Yotov's work is the introduction of heterogeneity that allows him to empirically measure these different flows and their relationship to each other. Several other papers have confirmed this, including [Bergstrand et al. \(2015\)](#), [Borchert and Yotov \(2017\)](#), and [Baier et al. \(2017\)](#). However, even with the domestic trade flow dummies, it is still possible to reproduce the distance puzzle in some specifications.

Adding the unobserved time-varying heterogeneity to the gravity model, as we propose in this paper, produces a distance parameter that naturally falls over time. Our solution to the distance puzzle is shown to be robust to changes in model specification, sample selection, and the trade data used. Moreover, the use of annual data allows us to track the reduction in distance elasticity over the past 50 years. Finally, in specifications that include intranational-time interactions, we are also able to measure a heterogeneous component that varies by country in addition to the changing border effect.

Overall, our estimation model manages to resolve several important biases in the estimates of trade growth. As a result, we are able to account for close to 70% of the change in trade for country pairs with at least 20 years of data. We find that the evolution in bilateral trade costs is driven mostly by distance elasticity, the EIAs and the change in border elasticity. The impact of the time-varying heterogeneity only has a small direct impact on the predicted change in growth, but its inclusion is of primary importance for removing the bias in the estimates of the other variables. Of course, an increased goodness-of-fit is to be expected since the model added a large number of variables. However, Bayesian model comparison tests (comparable to the Bayesian Information Criterion) strongly indicate that the improvement in the fit vastly outweighs the penalty for the increase in the number of estimated parameters.

² [Kwon et al. \(2022\)](#) also incorporates time-varying unobserved heterogeneity into their estimation of the impact of sanctions using the gravity model.

The remainder of the paper unfolds as follows. Section 2 delves deeper into unobserved heterogeneity, introducing a simple test for its detection. Sections 3 and 4 outline the estimation algorithm and the data used in the estimations. In Section 5, we analyze the estimated model parameters, and Section 6 sheds light on our estimated heterogeneity, coupled with an initial exploration of its likely determinants. Lastly, Section 7 synthesizes our findings, centering on the model's capacity to capture the dynamics of trade growth.

2. Time-varying heterogeneity and the gravity model

As the basis of our model, we make use of a general structural gravity model for panel data as outlined in Yotov et al. (2016) and Limão (2016):

$$X_{ij,t} = GS_{i,t}M_{j,t}\Psi_{ij,t} \quad (1)$$

In this equation, $X_{ij,t}$ represents the trade flows from exporter i to importer j at time t ; $S_{i,t}$ are the exporter's capabilities; $M_{j,t}$ captures the importer market characteristics; $\Psi_{ij,t}$ is a bilateral indicator of market access; and G is a gravitational constant.

In much of the gravity literature, the trade cost vector, $\Psi_{ij,t}$, takes the parametric form:

$$\Psi_{ij,t} = \exp\{z_{ij}\beta + w_{ij,t}\alpha + \phi_{ij}\} \quad (2)$$

where z_{ij} is a vector containing time-invariant variables that are plausibly exogenous, such as distance or contiguity, and β captures the impact of these variables on bilateral trade. The vector $w_{ij,t}$ contains the time-varying trade cost variables, which in this paper are economic integration agreements. Finally, ϕ_{ij} is a bilateral fixed effect that captures all time-invariant unobserved heterogeneity that may be correlated with time-varying variables of interest. Substituting this parameterized trade cost vector into Eq. (1) and using lowercase letters to indicate that the natural logs have been taken gives us the following equation:

$$x_{ij,t} = z_{ij}\beta + w_{ij,t}\alpha + \phi_{ij} + \delta_{i,t}^X + \delta_{j,t}^M + \epsilon_{ij,t} \quad (3)$$

Here, the dummy variables $\delta_{i,t}^X$ and $\delta_{j,t}^M$ capture the exporter-time and importer-time effects, respectively.

It is common to estimate this model using bilateral fixed effects. As is well known, the inclusion of such bilateral fixed effects does not allow us to estimate the coefficients on time-invariant variables. Further, if our interest is in the coefficient estimates on the time-varying variables ($w_{ij,t}$), fixed effects will only provide consistent estimates as long as the unobserved heterogeneity is constant or relatively slow-moving over time. Given the growth of trade and the expansion of free-trade areas over the last half century, it seems unlikely that the inclusion of bilateral fixed effects is a sufficient control for this endogeneity bias. We explore the plausibility of this assumption in the following subsection.

2.1. Testing for the presence of unobserved heterogeneity

The test employed to assess whether time-varying unobserved effects may be present is a variable addition test as proposed by Woolridge (2010). We start by rewriting the trade cost function as

$$\Psi_{ij,t} = \exp\{w_{ij,t}\alpha + \phi_{ij}\theta_t\} \quad (4)$$

assuming that the unobserved bilateral effect (ϕ_{ij}) interacts with a global process (θ_t). This process impacts all bilateral pairs, but each pair has a heterogeneous response.³ For simplicity, we normalize $\theta_1 = 1$ and omit the time-invariant variables z_{ij} . We substitute this trade cost function into the gravity Eq. (1) and take the logs to obtain the following expression:

$$x_{ij,t} = w_{ij,t}\alpha + \phi_{ij}\theta_t + \delta_{i,t}^X + \delta_{j,t}^M + \epsilon_{ij,t} \quad (5)$$

Following (Mundlak, 1978), we specify the relationship between the unobserved heterogeneity and the dependent variables as

$$\phi_{ij} = \bar{w}_{ij}\pi + \bar{\delta}_i^X\kappa + \bar{\delta}_j^M\nu + \mu_{ij} \quad (6)$$

where the bars above the variable indicate that its average has been computed over time for each bilateral pair. κ and ν are the coefficient vectors associated with averaged importer-year and exporter-year effects, and μ_{ij} is a bilateral random component uncorrelated with any of the right-hand side variables.⁴ We then substitute this expression for the bilateral fixed effects in Eq. (5). After rearranging, this yields the following.

$$\begin{aligned} x_{ij,t} &= w_{ij,t}\alpha + \delta_{i,t}^X + \delta_{j,t}^M + \theta_t \left(\bar{w}_{ij}\pi + \bar{\delta}_i^X\kappa + \bar{\delta}_j^M\nu + \mu_{ij} \right) + \epsilon_{ij,t} \\ &= w_{ij,t}\alpha + \delta_{i,t}^X + \delta_{j,t}^M + \bar{w}_{ij}\pi + \bar{\delta}_i^X\kappa + \bar{\delta}_j^M\nu + \left(\bar{w}_{ij}\pi + \bar{\delta}_i^X\kappa + \bar{\delta}_j^M\nu \right) + \lambda_t \left(\bar{w}_{ij}\pi + \bar{\delta}_i^X\kappa + \bar{\delta}_j^M\nu \right) + \zeta_{ij,t} \end{aligned} \quad (7)$$

where $\lambda_t = (\theta_t - 1)$. The composite error term $\zeta_{ij,t} \equiv \theta_t\mu_{ij} + \epsilon_{ij,t}$ is by construction uncorrelated with the other regressors.

Under the null hypothesis, the unobserved heterogeneity is constant over time ($\lambda_t = 0, \forall t = 2, \dots, T$). Eq. (7) is reduced to a linear equation that can be estimated using standard techniques.⁵ After estimating the model under the null hypothesis, the coefficient

³ Baier et al. (2014) and Bergstrand et al. (2015) assume that the global process evolves linearly over time; that is, globalization can be approximated by a linear time trend. Again, this may be appropriate for shorter time horizons, but seems less likely for longer panels.

⁴ Because the panel is unbalanced, we also need to take the time average of the exporter-year and importer-year dummy variables.

⁵ More specifically, we estimate the model including Mundlak formulation for the bilateral fixed effects and account for the exporter-year and importer-year fixed effects using the high-dimensional fixed effects technique developed by Guimarães and Portugal (2009).

estimates can be used to construct the estimated bilateral effect

$$\hat{\phi}_{ij} = \bar{w}_{ij} \hat{\pi} + \bar{\delta}_i^X \hat{\kappa} + \bar{\delta}_j^M \hat{\nu} \quad (8)$$

To perform the variable addition test, we then interact the constructed bilateral effect with time dummies for $t = 2, \dots, T$ and estimate the following equation:

$$x_{ij,t} = w_{ij,t} \alpha + \hat{\phi}_{ij} + \lambda_t \hat{\phi}_{ij} + \delta_{i,t}^X + \delta_{j,t}^M + error_{ij,t} \quad (9)$$

If the λ s differ from zero, the test provides evidence of time-varying unobserved heterogeneity. Furthermore, if the coefficients on $w_{ij,t}$ are sensitive to the inclusion of $\lambda_t \hat{\phi}_{ij}$, this may indicate that it is important to control for time-varying heterogeneity to obtain consistent estimates of the parameters of interest (Woolridge, 2010).

Table 1 displays the results of this variable addition test on a set of simple structural gravity models. Following the standard approach in the empirical gravity literature, we use five-year interval data, covering the period 1970–2015 when using COMTRADE (columns 1–4) and 1986–2016 when using ITPD-E (columns 5–8). We run the test on two specifications, each time comparing the coefficients of the models with time-varying heterogeneity (even columns) to those without (odd columns). The first specification (columns 1, 2, 5 and 6) summarizes the presence of EIAs using a single dummy variable, FTA^+ . The second specification (columns 3, 4, 7 and 8) splits up the EIA variable into six types of treaties.⁶ All models include bilateral, importer-year, and exporter-year fixed effects. More details on the data sources and control variables can be found in Section 4.

The even columns in Table 1 show the interactions on $\hat{\phi}$ to be statistically significant regardless of the specification or choice of trade data. This clearly suggests the presence of time variation in the unobserved heterogeneity. When comparing the OLS parameters to those in the variable addition test, we find a notable decrease in the coefficients of the EIAs (by as much as 50%). As $\hat{\phi}$ was computed under the null hypothesis, however, the parameters on the EIAs are still biased. Nevertheless, our findings do suggest that accounting for the time variation will likely have important ramifications for them. When we repeat the above exercise with a more general trade cost function (allowing for time-varying coefficients on the log of distance, contiguity, domestic trade, and language), the outcome of the test is the same. We reject the H_0 hypothesis of time-invariant heterogeneity and find considerable differences in parameter estimates.⁷ In summary, for all specifications, the tests indicate the presence of time-varying unobserved heterogeneity that is likely to affect our estimates.

2.2. Augmenting the gravity model

While the method described above provides a straightforward test for the presence of unobserved heterogeneity, it is only computationally feasible for relatively simple gravity models and small datasets. Furthermore, because the bilateral effect ($\hat{\phi}_{ij}$) was constructed under the null hypothesis, the estimates of α are not consistent (Woolridge, 2010). In order to allow for a more general structure that does not rely on the Mundlak assumption and can be estimated on more complex gravity models, we specify the following trade cost vector:

$$\Psi_{ij,t} = \exp\{w_{ij,t} \alpha + \gamma_{ij} + \phi_{ij} \theta_t\} \quad (10)$$

In this specification, we separate the time-varying unobserved heterogeneity into two components: the standard bilateral fixed effect (γ_{ij}) and the heterogeneous response to the unobserved global process that impacts the trade of countries around the world ($\phi_{ij} \theta_t$). To what extent this global process, θ_t , affects the trade of a specific country pair is determined by its elasticity of trade, ϕ_{ij} . The values of these elasticities are not restricted in any way. They can be both positive (increasing trade) and negative (decreasing trade). Most notably, they do not have to be symmetric, as ϕ_{ij} does not have to be equal to ϕ_{ji} .⁸

We further expand the model to allow for a more general trade cost specification that includes time-varying coefficients on the standard gravity variable and tracks the impact of economic integration agreements over time.

$$\Psi_{ij,t} = \exp\{z_{ij} \beta_t + w_{ij,t} \alpha_r + \gamma_{ij} + \phi_{ij} \theta_t\} \quad (11)$$

where β_t measures the impact of the time-invariant variables at time t and α_r measures the impact of an economic agreement in year r since its implementation. Substituting this trade cost function back into the gravity model (Eq. (1)) and adding the country-year fixed effects returns our main specification.

$$x_{ij,t} = z_{ij} \beta_t + w_{ij,t} \alpha_r + \gamma_{ij} + \phi_{ij} \theta_t + \delta_{i,t}^X + \delta_{j,t}^M + \epsilon_{ij,t} \quad (12)$$

This specification will be the focus of the remainder of this paper.

⁶ The six types of EIAs are non-reciprocal agreements, preferential agreements, free trade agreements, customs unions, common markets, or economic unions. FTA^+ combines the last four categories.

⁷ These results are available upon request.

⁸ In the trade literature, there have been others who have allowed for time-varying unobserved heterogeneity. The aforementioned papers by Baier et al. (2014) and Bergstrand et al. (2015) allow for unobserved effects that vary over time. In fact, the PPML package developed by Zylkin (2016) allows for an estimation that includes heterogeneous time trends. However, all of these specifications assume that the global process evolves linearly over time. In other areas, researchers have also stressed the importance of allowing for time-varying unobserved heterogeneity. For example, in the labor literature, it has been assumed that unobserved ability or talent can vary over time (Ding and Lehrer, 2014), and in the industrial organization literature, Olley and Pakes (1996) and Levinsohn and Petrin (2003) among others employ a control function approach to account for firm-specific productivity that can vary over time.

Table 1
Variable addition test for the presence of time-varying unobserved heterogeneity.

	COMTRADE				ITPD-E			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
FTA ⁺	0.358 ^a (13.29)	0.220 ^a (7.62)	–	–	0.349 ^a (10.98)	0.229 ^a (6.93)	–	–
NONRECIP	–	–	0.125 ^a (3.78)	0.106 ^b (3.21)	–	–	0.131 ^b (3.21)	0.0929 ^c (2.27)
PRFRNTL	–	–	0.214 ^a (7.01)	0.162 ^a (5.22)	–	–	0.0219 (0.60)	0.0266 (0.73)
FTA	–	–	0.380 ^a (12.41)	0.232 ^a (6.94)	–	–	0.342 ^a (9.60)	0.196 ^a (5.14)
CUNION	–	–	0.268 ^a (4.13)	0.124 (1.88)	–	–	0.384 ^a (4.94)	0.244 ^b (3.09)
COMMKT	–	–	1.132 ^a (18.21)	0.786 ^a (11.20)	–	–	0.989 ^a (13.67)	0.607 ^a (7.50)
ECON UNION	–	–	0.948 ^a (12.02)	0.584 ^a (6.81)	–	–	1.018 ^a (10.57)	0.606 ^a (5.85)
$\lambda_{1975}\hat{\phi}$	–	0.0363 ^c (2.12)	–	0.0381 ^c (2.15)	–	–	–	–
$\lambda_{1980}\hat{\phi}$	–	0.101 ^a (5.92)	–	0.103 ^a (5.80)	–	–	–	–
$\lambda_{1985}\hat{\phi}$	–	0.0984 ^a (5.68)	–	0.0972 ^a (5.40)	–	–	–	–
$\lambda_{1990}\hat{\phi}$	–	0.0919 ^a (5.45)	–	0.0878 ^a (5.02)	–	0.125 ^a (5.61)	–	0.141 ^a (5.58)
$\lambda_{1995}\hat{\phi}$	–	0.107 ^a (6.68)	–	0.0940 ^a (5.66)	–	0.145 ^a (6.77)	–	0.156 ^a (6.37)
$\lambda_{2000}\hat{\phi}$	–	0.161 ^a (10.19)	–	0.147 ^a (8.97)	–	0.204 ^a (9.55)	–	0.212 ^a (8.70)
$\lambda_{2005}\hat{\phi}$	–	0.188 ^a (11.95)	–	0.165 ^a (10.10)	–	0.229 ^a (10.74)	–	0.232 ^a (9.52)
$\lambda_{2010}\hat{\phi}$	–	0.175 ^a (11.13)	–	0.155 ^a (9.54)	–	0.237 ^a (11.07)	–	0.238 ^a (9.72)
$\lambda_{2015}\hat{\phi}$	–	0.140 ^a (8.93)	–	0.119 ^a (7.30)	–	0.197 ^a (9.21)	–	0.195 ^a (7.97)
Constant	13.88 ^a (3143.09)	15.10 ^a (120.56)	13.84 ^a (1859.94)	14.98 ^a (110.38)	0.610 ^a (111.62)	1.866 ^a (14.68)	0.575 ^a (56.30)	1.893 ^a (12.55)
FE panel	yes	–	yes	–	yes	–	yes	–
FE imp-year	yes	yes	yes	yes	yes	yes	yes	yes
FE exp-year	yes	yes	yes	yes	yes	yes	yes	yes
Observations	181,579	181,579	181,579	181,579	129,490	129,490	129,490	129,490
R ²	85.7%	85.8%	85.8%	85.8%	88.0%	88.0%	88.0%	88.0%

Variable addition test run on a gravity model using both COMTRADE (columns 1–4) and ITPD-E trade data (columns 5–8). Odd columns are the standard OLS results, even columns include the Mundlak device, $\hat{\phi}$, obtained from the first-stage regressions. For the ITPD-E regressions, the interaction terms λ_t refer to the $(t+1)$ th year. T-values in parentheses. ^a, ^b, ^c indicates significance at the 0.1%, 1% and 5% level.

3. Estimation strategy

The model described in Eq. (12) is a straightforward log-linear gravity model with high-dimensional fixed effects, augmented with unobserved heterogeneity as outlined above. This addition adds a large number of parameters to the model. While θ_t only has as many free parameters as there are years, ϕ_{ij} has to be estimated for each importer–exporter combination (more than 30,000 in the case of this paper).

Inspired by Guimarães and Portugal (2009) we simplify the estimation problem by separating the estimation of the unobserved heterogeneity from the other variables in the model. We do this using a Bayesian Gibbs sampling algorithm: we split up the estimation problem into a number of simple conditional density functions and iteratively draw from them using a Monte Carlo Markov chain algorithm. After a sufficient burn-in⁹ has been completed, the resulting draws can be used to reconstruct the distribution of the parameters of the model.

More specifically, if the unobserved heterogeneity is known, the remaining parameters (α_r , β_t , $\delta_{i,t}^M$, $\delta_{j,t}^X$, γ_{ij} and σ^2) can be estimated using standard gravity model estimation techniques. Vice versa, if the latter variables are known, the problem of estimating the unobserved global process (θ_t) and bilateral trade elasticities (ϕ_{ij}) is reduced to identifying a single underlying factor from a panel of correlated variables, $\omega_{ij,t}$:

$$\omega_{ij,t} \equiv X_{ij,t} - z_{ij}\beta_t - w_{ij,t}\alpha_r - \delta_{i,t}^M - \delta_{j,t}^X - \gamma_{ij}$$

⁹ The burn-in are the first iterations of the estimator that have not yet converged to the posterior density.

$$= \phi_{ij}\theta_t + \epsilon_{ij,t} \quad (13)$$

If N denotes the number of countries, we can stack the cross sections of $\omega_{ij,t}$, ϕ_{ij} and $\epsilon_{ij,t}$ into $(N^2 \times 1)$ vectors Ω_t , ϕ and ϵ_t , where the lack of subscript indicates that the variables have been stacked.¹⁰ This allows us to rewrite the problem of generating θ_t as a straightforward factor analysis:

$$\Omega_t = \phi \theta_t + \epsilon_t \quad (14)$$

$$\epsilon_t \sim N(0, \sigma^2 I) \quad (15)$$

In theory, it should then be possible to identify the unobserved global process using a principal component analysis (PCA). However, like most panel datasets of bilateral trade, our dataset is unbalanced: the trade data of about one in four country pairs does not cover the entire period. While the PCA can be adjusted to handle missing observations (e.g., by using an alternating least-squares algorithm), the size of the dataset in combination with the high number of missing trade data makes the PCA algorithm much too slow.

To solve this problem, we go one step further, rewriting the problem of identifying θ_t as a Gaussian state-space model. This allows us to use Durbin and Koopman's (2012) univariate treatment of the Kalman filter, designed to estimate the distribution of unknown factors from very large (state-space) models. The Kalman filter also has an integrated solution to unbalanced panel data in that it replaces the missing observations with meaningless noise. As such, the observations are no longer missing; they just do not hold any informational value. The net result is that the computing time of θ_t in each iteration of the Gibbs sampler is reduced from over half an hour to less than a second, enabling us to run the thousands of iterations that the algorithm requires.¹¹

To construct this state-space model, we have to be explicit about the temporal behavior of our unknown (state) variable, θ_t . Typically, a state-space model imposes an autoregressive pattern with one lag (cf. Kim and Nelson, 1999). As we want to allow the data itself to determine what, if any, pattern is present, we instead introduce the more simple assumption that θ_t is drawn from a normal distribution. As such, any patterns in θ_t will only be informed by the trade data and not our model assumptions. Combined with the identifying assumptions outlined above, this gives us the following state equation:

$$\theta_t \sim N(0, 1) \quad (16)$$

Combining this with Eq. (14) completes the state-space model.

In summary, each iteration of the Gibbs sampling algorithm loops repeatedly through the steps outlined below. Each step uses the values of the parameters drawn in the previous steps, but for the sake of notational simplicity, we omit the superscript indicating the iteration.

1. Conditional on $\phi_{ij}\theta_t$, compute and draw values of $\alpha_r, \beta_t, \gamma_{ij}, \delta_{i,t}^M, \delta_{j,t}^X$ and σ^2 .

- (a) Draw values of α_r and β_t conditional on $\phi_{ij}\theta_t, \gamma_{ij}, \delta_{i,t}^M, \delta_{j,t}^X$ and σ^2 from a normal distribution.
- (b) Compute the fixed effects $\gamma_{ij}, \delta_{i,t}^M$ and $\delta_{j,t}^X$.
- (c) Draw values of σ^2 from an inverse Wishart distribution.

2. Conditional on $\alpha_r, \beta_t, \gamma_{ij}, \delta_{i,t}^M, \delta_{j,t}^X$ and σ^2 , compute and draw values of ϕ_{ij} and θ_t .

- (a) Compute $\omega_{ij,t} \equiv \phi_{ij}\theta_t + \epsilon_{ij,t}$ and stack this into a $(N^2 \times 1)$ matrix Ω_t
- (b) Combine Eq. (14) with Eq. (16) into the following state-space model and use the univariate treatment of the Kalman filter of Durbin and Koopman (2012) to draw new values for θ_t :

$$\Omega_t = \phi \theta_t + \epsilon_t \quad (17)$$

$$\theta_t \sim N(0, 1) \quad (18)$$

- (c) Conditional on θ and Ω , draw new values for ϕ_{ij} from a normal distribution.

More details on the estimation algorithm, including the exact conditional posterior probabilities, can be found in Appendix B.

As our model is estimated using Bayesian econometrics, we have to be explicit about our prior assumptions. For the most part, we used actual but uninformative priors by specifying a (natural conjugate) distribution with such a high variance that it does not impose any meaningful restrictions on the estimated parameters.

$$\begin{aligned} \beta_t &\sim N(\mu_{\beta_t}, V_{\beta_t}) & \delta_{it}^M &\sim N(\mu_M, V_M) \\ \alpha_r &\sim N(\mu_{\alpha_r}, V_{\alpha_r}) & \delta_{jt}^X &\sim N(\mu_X, V_X) \\ \sigma^2 &\sim iWish(S_\sigma, \nu_\sigma) \end{aligned} \quad (19)$$

¹⁰ E.g., ϕ is a $(N^2 \times 1)$ vector defined as $\phi = [\phi_{11}, \dots, \phi_{1n}, \phi_{21}, \dots, \phi_{nn}]'$.

¹¹ We assessed the algorithm's capability to extract a shared heterogeneous process through a Monte Carlo simulation. Additionally, we demonstrated that any bias in the Ordinary Least Squares (OLS) results arises only when there is a correlation between unobserved heterogeneity and the explanatory variables. In particular, our estimation approach consistently identified the overarching trend, irrespective of this correlation. Further information regarding the simulation exercises can be provided upon request.

where $iWish$ is the inverted Wishart distribution.

However, we did impose an additional structure on those variables γ_{ij} and ϕ_{ij} , that differ between country pairs, as the number of years available to determine the values of these parameters is limited.¹² Specifically, we used a hierarchical prior, a common solution to this large-N small-T problem in Bayesian panels.¹³ While the value of each ϕ_{ij} is still uncorrelated over country pairs, the assumption that they come from the same underlying distribution can now be used to improve their estimation. Specifically, γ_{ij} and ϕ_{ij} are modeled as i.i.d. draws from a normal distribution:

$$\begin{aligned}\gamma_{ij} &\stackrel{iid}{\sim} N(\mu_\gamma, V_\gamma) \\ \phi_{ij} &\stackrel{iid}{\sim} N(\mu_\phi, V_\phi)\end{aligned}\quad (20)$$

Unlike the prior assumptions on, e.g., β_t , the values of the parameters $\mu_\gamma, \mu_\phi, V_\gamma$ and V_ϕ are estimated within the model. They, in turn, have the following (actual but uninformative) prior distributions:

$$\begin{aligned}\mu_\gamma &\sim N(\mu_{\mu_\gamma}, V_{\mu_\gamma}) & V_\gamma &\sim iWish(S_{V_\gamma}, v_{V_\gamma}) \\ \mu_\phi &\sim N(\mu_{\mu_\phi}, V_{\mu_\phi}) & V_\phi &\sim iWish(S_{V_\phi}, v_{V_\phi})\end{aligned}\quad (21)$$

4. Data

We use two different trade datasets to present our results: the COMTRADE dataset, which covers 256 countries from 1970 to 2019, and the International Trade and Production Database for Estimation (ITPD-E version 2), which includes 252 countries that span the period from 1986 to 2019.¹⁴ A shortcoming of the COMTRADE dataset is that it does not include data on intranational trade, which we therefore approximate by subtracting total exports from GDP. The ITPD-E database, on the other hand, includes measurements of intranational trade that are carefully constructed and theoretically consistent. However, it has a shorter time horizon, which may yield less precise estimates of the ϕ_{ij} s.

In order to differentiate intra- from international trade, we include domestic border dummies as a regressor (*domestic*). As our model also includes bilateral fixed effects, we interact these domestic barrier dummies with time to account for the globalization process, as suggested by Yotov (2012) and Bergstrand et al. (2015). Data on expenditure-based GDP come from the Penn World Tables. The remaining gravity variables, such as the log of population-weighted distance (*lnDistw*), contiguity, or common ethnographic language (*language*), come from the CEPII gravity dataset (Conte et al., 2022). As with the domestic border dummies, we interact these variables with time (Bergstrand et al., 2015). In each case, the most recent year was omitted to avoid problems of perfect multicollinearity. As such, the coefficient on these variables should be interpreted as the change with respect to 2019.

Data on economic integration agreements come from the July 2021 edition of the EIA database, constructed by Jeffrey Bergstrand and Scott Baier.¹⁵ Here, we include dummy variables indicating the existence of a non-reciprocal agreement (NONRECIP), a preferential trade agreement (PREFRNTL), a free trade agreement (FTA), a customs union (CUNION), a common market (COMMKT), or an economic union (ECON UNION). We also included a dummy, FTA^+ , indicating the existence of either a FTA, CUNION, COMMKT or ECON UNION. All dummies were interacted with time to track how long both countries have been members of the same agreement on a yearly basis. The regressions also contain a dummy variable that indicates whether an agreement will be signed in the next five years. As the number of country pairs with a long-term agreement steadily decreases as we increase the age of the agreement, we capped the examination of their impact to 25 years after the agreement came into effect. The final coefficient tracks the impact after 25 years or more. We reduced the period of analysis to 20 years for common markets and 15 years for economic unions, as there have been fewer country pairs with these agreements for this long.

5. Results

The model outlined in this paper allows us to estimate a set of gravity models that control for and identify the presence of time-varying unobserved heterogeneity. All models are estimated using yearly data for all country pairs and include importer-year, exporter-year, and bilateral fixed effects. We will focus our discussion on one preferred specification, which examines the yearly impact of each type of EIA separately while controlling for distance, contiguity, common language, and domestic border (all interacted with time).

Overall, our estimates prove to be remarkably stable across all specifications and between datasets, as illustrated in Tables 2 and 3.¹⁶ In both tables, column 1 starts with a model estimated using the COMTRADE data that, apart from the unobserved heterogeneity, only includes the EIA dummies. Column 2 adds the population-weighted distance; and column 3 complements this with common language, contiguity, and the domestic border dummies. Column 4 repeats this last specification, but without the correction for unobserved heterogeneity. Columns 5 through 8 duplicate this analysis with the ITPD-E dataset.

¹² At most, there are 45 observations for each country pair, but due to missing data, more than three-quarters of the country pairs have fewer.

¹³ Other names this approach is associated with are random coefficients model or empirical Bayes method.

¹⁴ The ITPD-E dataset is available at <https://www.usitc.gov/data/gravity/itpde.htm>. More details on its properties and construction can be found in Borchert et al. (2021, 2022).

¹⁵ This database, available at <https://www3.nd.edu/~jbergstr/>, only covers EIAs up to the year 2017. Any changes in membership or new agreements in 2018 and 2019 were filled in using the information from Miller and Standaert (2023).

¹⁶ These results are similar to Campos et al. (2021) who compare the three most common approaches to measuring domestic trade and show that they lead to very similar estimates of the parameters that are usually estimated within a structural gravity framework. Bergstrand et al. (2015) find similar results.

Table 2
Baseline Results with Aggregated Economic Integration Agreement.

Variables	COMTRADE				ITPD-E			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
FTA or more	0.540 ^a (0.016)	0.370 ^a (0.016)	0.344 ^a (0.014)	0.328 ^a (0.012)	0.630 ^a (0.014)	0.438 ^a (0.015)	0.393 ^a (0.015)	0.301 ^a (0.014)
$\phi_{ij}\theta_i$	yes	yes	yes	–	yes	yes	yes	–
ln(Distw) × time	–	yes	yes	yes	–	yes	yes	yes
Contiguity × time	–	–	yes	yes	–	–	yes	yes
Domestic × time	–	–	yes	yes	–	–	yes	yes
Language × time	–	–	yes	yes	–	–	yes	yes
FE panel	yes	yes	yes	yes	yes	yes	yes	yes
FE isoim_year	yes	yes	yes	yes	yes	yes	yes	yes
FE isoex_year	yes	yes	yes	yes	yes	yes	yes	yes
Observations	1,129,337	957,576	957,576	964,817	821,571	681,649	681,649	686,274
R ²	87.6%	87.8%	87.8%	85.0%	88.5%	88.6%	88.6%	86.1%

Estimation of the gravity model using the COMTRADE (columns 1–4) and ITPD-E datasets (columns 5–8). Columns 1–3 and 5–7 include the heterogeneous response to globalization and are estimated using the Kalman filter. Columns 4 and 8 are estimated on the same dataset using Stata's reghdfe program. Standard errors are in parentheses. ^a, ^b, ^c indicates significance at the 0.1%, 1% and 5% level.

Table 3
Baseline Results with Economic Integration Agreement by Type.

Variables	COMTRADE				ITPD-E			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
NONRECIP	0.350 ^a (0.014)	0.144 ^a (0.015)	0.138 ^a (0.015)	0.142 ^a (0.014)	0.331 ^a (0.018)	0.116 ^a (0.019)	0.122 ^a (0.018)	0.130 ^a (0.017)
PREFRNTL	0.311 ^a (0.015)	0.105 ^a (0.014)	0.103 ^a (0.015)	0.196 ^a (0.013)	0.286 ^a (0.020)	0.144 ^a (0.019)	0.134 ^a (0.020)	0.050 ^a (0.016)
FTA	0.642 ^a (0.016)	0.403 ^a (0.016)	0.378 ^a (0.016)	0.364 ^a (0.013)	0.691 ^a (0.018)	0.442 ^a (0.018)	0.400 ^a (0.019)	0.311 ^a (0.016)
CUNION	0.726 ^a (0.030)	0.424 ^a (0.029)	0.398 ^a (0.030)	0.278 ^a (0.027)	1.051 ^a (0.037)	0.749 ^a (0.034)	0.696 ^a (0.036)	0.340 ^a (0.033)
COMMKT	1.168 ^a (0.032)	0.893 ^a (0.034)	0.848 ^a (0.033)	1.069 ^a (0.027)	1.269 ^a (0.036)	0.975 ^a (0.036)	0.893 ^a (0.037)	0.967 ^a (0.032)
ECON UNION	1.339 ^a (0.047)	0.972 ^a (0.049)	0.915 ^a (0.045)	0.909 ^a (0.035)	1.508 ^a (0.050)	1.182 ^a (0.053)	1.064 ^a (0.054)	1.032 ^a (0.042)
$\phi_{ij}\theta_i$	yes	yes	yes	–	yes	yes	yes	–
ln(Distw) × time	–	yes	yes	yes	–	yes	yes	yes
Contiguity × time	–	–	yes	yes	–	–	yes	yes
Domestic × time	–	–	yes	yes	–	–	yes	yes
Language × time	–	–	yes	yes	–	–	yes	yes
FE panel	yes	yes	yes	yes	yes	yes	yes	yes
FE isoim_year	yes	yes	yes	yes	yes	yes	yes	yes
FE isoex_year	yes	yes	yes	yes	yes	yes	yes	yes
Observations	1,129,337	957,576	957,576	964,274	821,571	681,649	681,649	686,274
R ²	87.6%	87.8%	87.8%	85.0%	88.5%	88.6%	88.6%	86.1%

Estimation of the gravity model using the COMTRADE (columns 1–4) and ITPD-E datasets (columns 5–8). Columns 1–3 and 5–7 include the heterogeneous response to globalization and are estimated using the Kalman filter. Columns 4 and 8 are estimated on the same dataset using Stata's reghdfe program. Standard errors are in parentheses. ^a, ^b, ^c indicates significance at the 0.1%, 1% and 5% level.

When we compare our results to the standard OLS estimates, we see that controlling for unobserved heterogeneity has a strong impact on the EIA variables provided that they are considered separately or that the ITPD-E data is used.¹⁷ In the latter case, the impact of signing an FTA increases from 35% ($= e^{0.301} - 1$) to 48% ($= e^{0.393} - 1$). Our estimation model also reduces the discrepancy between the COMTRADE and ITPD-E coefficients. For example, in Table 3, it halves the impact of a preferential agreement when using COMTRADE (from 22% to 11%) and triples it when using ITPD-E (from 5% to 14%), making the difference between both datasets statistically insignificant. Controlling for unobserved heterogeneity also increases the estimated impact of economic unions, showing them to exceed the impact of common markets in the COMTRADE dataset, which mimics the pattern in the ITPD-E results. Despite this convergence, the ITPD-E estimates remain slightly higher than those using the COMTRADE data (about 10% to 20%),

¹⁷ The one specification where we see only a limited impact on the FTA⁺ variable is when the COMTRADE data is used (columns 3 and 4, Table 2).

a difference that persists when both datasets are restricted to the same sample.¹⁸ Therefore, this is likely driven by the improved domestic trade data of ITPD-E.

In both tables, the incremental incorporation of the time-invariant control variables moderates the impact of the EIAs. This is most noticeable when we add the distance parameters (column 2), which drops the impact of EIAs by a third to half, depending on the agreement type and dataset used. The severity of this decrease alludes to one of the strongest results we find in our paper, namely the impact of controlling for unobserved heterogeneity on the distance parameter. We return to the time-varying parameter estimates on distance in Section 5.2.

5.1. Measuring the impact of economic integration agreements over time

In order to account for the dynamic impact of EIAs on trade flows, we resorted to the now common practice of using leads and lags, as suggested by Baier and Bergstrand (2007). Less typical, however, is our use of annual data. Since (Cheng and Wall, 2005), most research has used interval data to capture these dynamics. As pointed out by these authors, the dependent and independent variables may not be able to fully adjust within one year, which could lead to noisy or biased coefficient estimates. Baier et al. (2014) attempted to work around this issue by using a model that imposes linear trends in the impact of EIAs.¹⁹ More recently, Egger et al. (2022) have noted that due to the timing of recent trade agreements, using interval data may result in downward-biased coefficient estimates. They argue that careful estimation using annual data can provide better insights into the dynamic impacts of trade agreements on trade flows.²⁰

Expanding on the analysis of Egger et al. (2022), we track the impact of different types of EIAs on a yearly basis. Because we control for unobserved heterogeneity, our estimation method filters out some of the noise and corrects for potential endogeneity caused by changes in the overall economic conditions. As we show, this significantly increases the precision of the yearly EIA estimates. Furthermore, unlike Baier et al. (2014), we do not have to impose any trends and can track the impact of the agreements for up to 25 years. Figs. 1 and 2 present the estimated coefficients of the different EIAs over time using our preferred specification on the COMTRADE and ITPD-E datasets. In addition to the parameter estimates (crosses), the vertical lines indicate the 95% confidence intervals, while the red line contrasts this with the regular OLS estimation results.

Before we discuss the impact of specific EIAs, several observations are noteworthy. First, comparing our results to the OLS estimates, we see that the impact of FTAs, customs unions, common markets, and economic unions is quite strongly underestimated when unobserved heterogeneity is not accounted for. This is particularly notable in panel d of Fig. 2, where the impact of customs unions fluctuates around zero in the OLS results. Second, the coefficient estimates also tend to be more stable, particularly in the case of the COMTRADE data. See, e.g., the parameters on the FTAs (panel c, Fig. 1). Third, accounting for unobserved heterogeneity also ensures a better alignment between the EIA coefficients with and without time interactions.²¹ For example, the average OLS coefficient on economic unions is 0.9 in Table 3, even though the yearly estimates never exceed 0.7 in panel f of Fig. 1.

In line with the literature, Figs. 1 and 2 show different phases in the impact of EIAs, although these differ quite strongly depending on the type of agreement. For non-reciprocal agreements, the trade-creating effects increase over the first 8 to 10 years, then level off around year 10. For preferential trade agreements, we find that, on average, bilateral trade is about 16% higher before the agreement's implementation compared to bilateral pairs without an agreement. This increases to about 40% higher around year 10 and starts to decline around year 16 of the agreement.²² For free trade agreements, the trade-creating effects follow a similar path to the one discussed in Baier et al. (2014) and Egger et al. (2022). In the years before the agreement, trade was higher than predicted. When the agreement goes into effect, there is an immediate impact, followed by a growth phase over the agreement's first 6 to 8 years, when it reaches the mature stage. Customs unions follow a similar, albeit more noisy, path. For common markets and economic unions, the immediate impact is considerable, exceeding that of the other agreements. This is likely related to the existence of another type of agreement beforehand. In contrast, their growth phase appears to be relatively modest, lasting around 4 to 5 years.

5.2. Globalization and the diminishing role of distance

The improvement in our estimation model is not limited to the EIA coefficients. One of the most notable changes is centered on distance. Fig. 3 depicts the distance elasticity of trade, with (black) and without (red) correcting for unobserved heterogeneity. Both models are estimated with the preferred specification, which includes the domestic trade dummies.

¹⁸ Results are available upon request.

¹⁹ In particular, they split up the dynamic impact of an EIA into a pre-economic integration agreement trend, an impact effect, and a post-economic integration agreement trend. The authors found that, for most types of agreements, there was a statistically significant impact effect at the time of the agreement and trade flows increased over the next 10 to 15 years.

²⁰ In estimating the effects of free trade agreements, defined as *FTA*⁺ in this paper, they document several phases: (1) the pre-agreement anticipation phase, (2) a growth phase, which covers the first few years the agreement is in force and (3) a mature phase where the impact of the agreement on trade flows has reached its full effect—around eight years after implementation.

²¹ Specifically, the coefficients in Tables 2 and 3 should equal a weighted average over the entire time period, with the weights favoring the initial observations, as the number of country pairs with longer running agreements steadily decreases.

²² This decline may be due to a smaller sample size as there are a limited number of country pairs with agreements that have been in effect for more than 15 years.

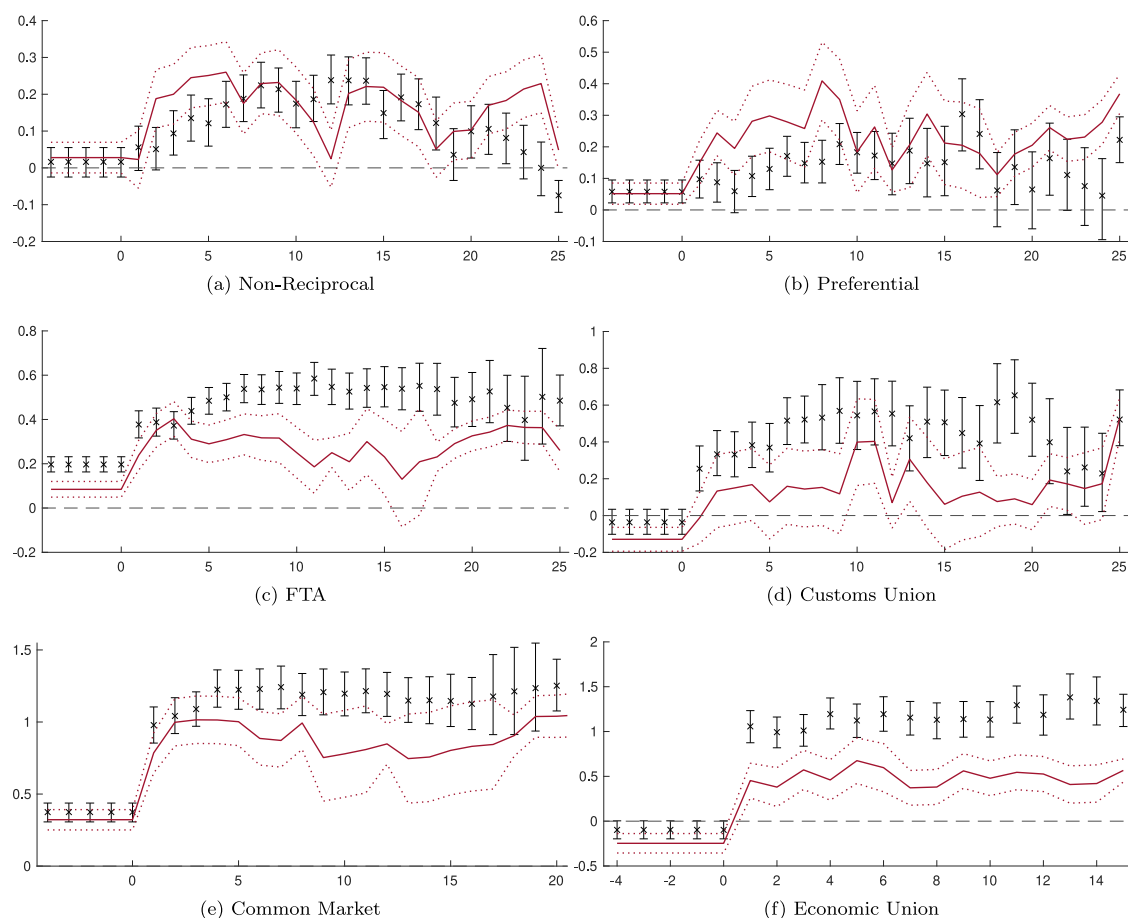


Fig. 1. EIA parameters over time using the COMTRADE data.

NOTES: EIA estimates (black Xs) and 95% confidence intervals (vertical lines) when correcting for unobserved heterogeneity and using the preferred specification. The uncorrected OLS estimates and their 95% confidence interval are drawn using the red (dotted) lines. Note that the scales of the x and y -axes differ between the panels.

As these distance parameters are relative to the final year in the dataset (2019), we expect the coefficients to converge to zero. We also expect the parameter values to start out negative and gradually increase, indicating the increasing irrelevance of distance as a determinant of international trade. This is not what we see in the OLS estimates, which on both datasets decrease over time, thus reproducing the infamous distance puzzle.²³ When we account for unobserved heterogeneity, we instead see a slowly increasing parameter that converges to zero, consistent with the idea of globalization as leading to the “death of distance” (Cairncross, 1997).²⁴ The absolute value of the distance parameter decreases by around 0.4, implying an annual growth in trade of 3.9% for a country pair that is separated by the median distance.²⁵ All international trade flows see this improvement, most of which are of similar size: 90% of country pairs have an implied yearly growth rate of 3.2% to 4.2%.

5.3. The domestic border and other control variables

The remaining time-invariant regressors turn out to be less relevant. Only the initial coefficients (before the early 1990s) are significant for contiguity and common language. Omitting these variables has no meaningful impact on the coefficient estimates of any other variable, including the identification of the unobserved heterogeneity (cf. Section 6).

The domestic trade dummies fare slightly better, although they still become insignificant from 2005 onward for the ITPD-E data (cf. Fig. 4). Surprisingly, the sign of these coefficients runs counter to our expectations. As suggested by Yotov (2012), we would

²³ In the COMTRADE data, the OLS estimates do not converge towards zero, implying a sudden jump in the effect of distance in 2019.

²⁴ Estimating the model using PPML also resolves the distance puzzle but results in mostly insignificant distance parameters from 1986 onward.

²⁵ The median distance (logged) is 8.83, which means an increase in the log of trade of $0.4 \times 8.83 = 3.53$ over 33 years, which comes down to a yearly growth of $3.53^{1/33} = 1.0389$ or 3.9%.

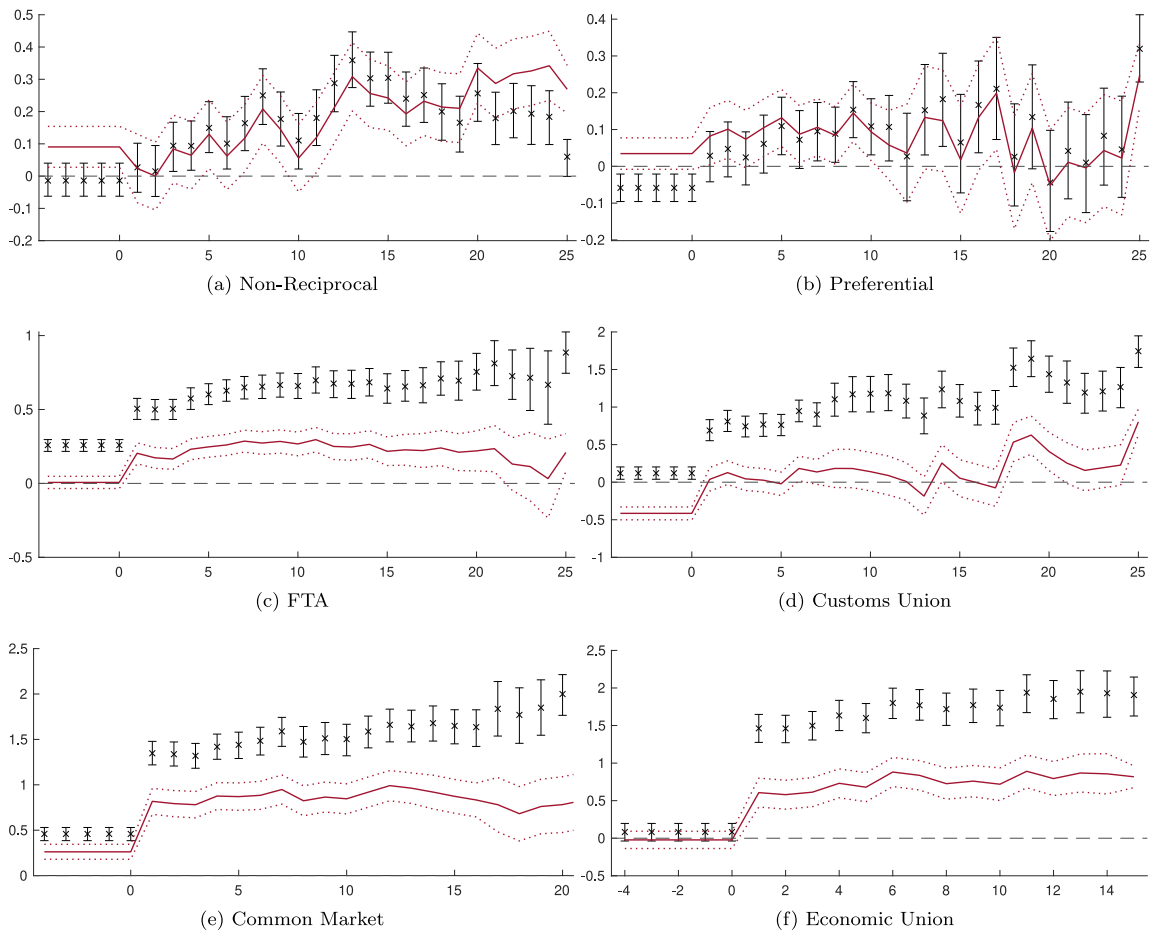


Fig. 2. EIA parameters over time using ITPD-E data.

NOTES: EIA estimates (black Xs) and 95% confidence intervals (vertical lines) when correcting for unobserved heterogeneity and using the preferred specification. The uncorrected OLS estimates and their 95% confidence interval are drawn using the red (dotted) lines. Note that the scales of the x and y-axes differ between the panels.



Fig. 3. Distance parameter over time.

NOTES: Parameter estimates of the distance dummies (black Xs) and 95% confidence intervals (vertical lines) when correcting for unobserved heterogeneity and using the preferred specification. The uncorrected OLS estimates and their 95% confidence interval are drawn using the red (dotted) lines.

expect the coefficients on the domestic border dummies to start positively and slowly decrease over time as globalization pushes countries toward international trade. Yet, both the COMTRADE and ITPD-E estimates follow the opposite pattern, starting below zero and increasing over time. Nevertheless, considered within the wider framework of the model, there are two reasons why our results are still consistent with the globalization pattern established in the literature.

First, the domestic trade dummies are not the only parameters that capture the change over time in domestic trade. In contrast to traditional gravity models, the intranational trade flows are also affected by the unobserved heterogeneity in our estimation. As

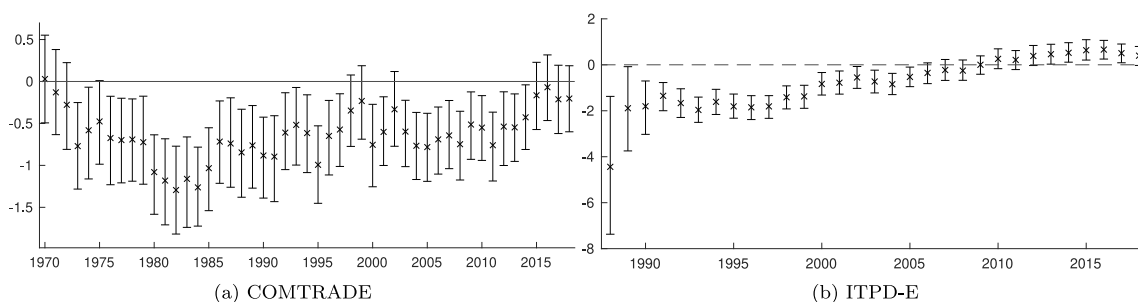


Fig. 4. Domestic border dummy over time.

NOTES: Parameter estimates of the border dummies and their 95% confidence interval (vertical lines) using the preferred model specification.

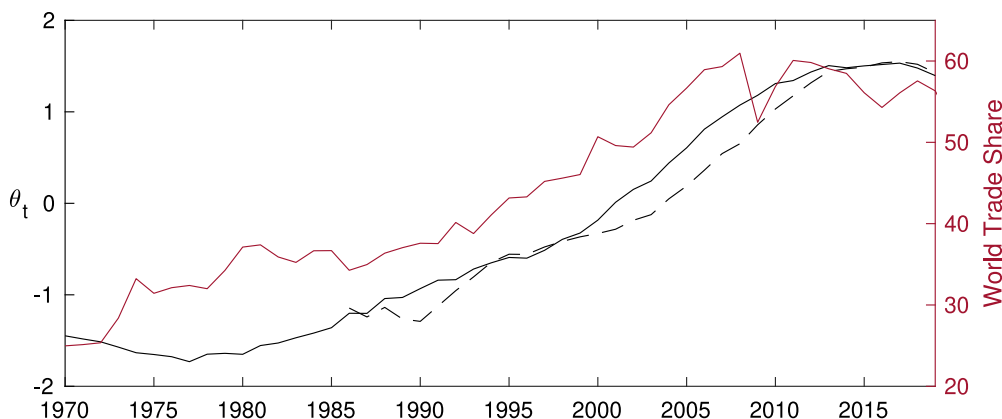


Fig. 5. World trade shares vs. the unobserved heterogeneity.

NOTES: The estimates of θ_t are represented by the black uninterrupted line (COMTRADE) and dashed line (ITPD-E) and plotted along the left y-axis. The world trade shares (source: Worl Bank) are plotted along the right y-axis in red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

we show in the following section, the own-trade elasticity, ϕ_{ii} , is negatively associated with domestic trade. This correlation is of sufficient size to compensate for and even reverse the increase in domestic trade suggested in Fig. 4.²⁶ Second, a large part of the globalization process is already captured by the improvement in the distance parameter estimates. As pointed out in the previous section, the decrease in the distance coefficient has a similar effect on most country pairs, with the exception of domestic trade flows that have a (logged) distance of zero.

6. What is behind the unobserved heterogeneity?

6.1. The global trend in unobserved heterogeneity

In light of the influence the unobserved heterogeneity exerts on the estimates of the gravity model, it is inevitable to contemplate its constituting factors. To start answering that question, Fig. 5 plots the values of the global process, θ_t , for both the COMTRADE (black full line) and ITPD-E datasets (black dashed line). Both estimates are highly correlated over time.²⁷ As the estimation procedure was set up not to assume any time-dependency (cf. Eq. (16)), this random walk in the global process is clearly a feature of the data and not imposed by the estimation itself. What we did impose as an identifying assumption on θ_t is its average value of zero and standard deviation of one.

Fig. 5 shows a strong increase in θ_t , particularly in the early 1990s, leveling off in the mid-2000s. This pattern turns out to be remarkably stable. First, none of the changes in the model specification that have been discussed so far, including those in Tables 2 and 3, result in any noticeable difference. As the estimates of θ_t contain a unit root, a straightforward correlation risks overestimating the correspondence. However, even the correlation in the growth of θ_t between specifications never drops below

²⁶ When we estimate a specification without the domestic border dummies, the specification still accounts for the globalization process described by Yotov (2012) as the ϕ_{ii} s adjust to fill its role. As such, the impact of omitting these dummies remains relatively limited. Results are available upon request.

²⁷ The Dickey-Fuller test confirms the presence of a unit root in the estimates of both datasets.

98.4%. Second, switching between the COMTRADE and ITPD-E data has a larger impact, decreasing the correlation in the growth of θ_t to 59.7% (Fig. 5). This is due, at least in part, to sample selection effects. When we restrict the sample to observations covered by both COMTRADE and ITPD-E, the correlation is brought back up to 79.1%. Third, running the regression with only those dyads that remain present throughout the entire dataset results in a similar pattern.²⁸ The Appendix A provides an overview of a number of additional robustness checks that we ran.

Given its similarity to the globalization process described at the beginning of this paper, we contrasted the plots of θ_t with the share of global trade relative to global production. The red line in Fig. 5 shows world trade shares, the values of which can be read off the right-hand y-axis. From the 1970s onward, these increased from about 25% to almost 60% up until the financial crisis in 2008. Although the resulting drop in trade was quickly offset, the period of trade growth did not return. While in general, θ_t appears to follow the time path of global trade shares, it is difficult to directly link it to a specific factor in the overall globalization pattern. Plausible candidates include steadily falling transportation and information costs, containerization, lower tariffs and non-tariff barriers, and global value chains (Feenstra, 1998; Krugman, 1995). While there are not enough observations in θ_t to identify which (if any) of these processes are driving it, we can leverage the information in its elasticity, ϕ_{ij} , to establish some likely correlations.

6.2. The trade elasticity to the global trend

Before delving into the analysis of ϕ_{ij} , we take a moment to examine its general shape and structure. Unlike θ_t , we did not impose any restrictions on ϕ_{ij} . The fact that their values are centered on zero is due to the inclusion of the importer-year, exporter-year, and bilateral fixed effects. The elasticities follow a normal distribution, with 90% of the estimates lying between -1 and 1 . Combined with θ_t , this corresponds to a change in the trade cost variable by a factor of 2.4 for COMTRADE and 2.6 for ITPD-E. Only a third of the ϕ_{ij} s are significant at the 5% level. Like the values of θ_t , their estimates are robust to the changes in the specification listed above.²⁹ The biggest change occurs again when we switch from one dataset to the other. Fig. 6 compares the values of ϕ_{ij} and θ_t between the two. While the overall correlation is only 64.2%, much of the difference is due to sample selection effects, as restricting the sample to observations where both datasets are available increases the correlation to 81.3%.

The large number of parameters in ϕ_{ij} provides an opportunity to identify some of the factors behind the unobserved heterogeneity. To this end, we ran several regressions, presented in Table 4 below, performing two sets of tests. First, they verify whether and how the unobserved heterogeneity is correlated with the explanatory variables. To this end, the time-invariant variables were included as is; for the EIA variables, we included a dummy indicating whether the country pair signed an agreement during the period of study. Second, the regressions explore the underlying factors that contribute to the heterogeneity. We gathered data on several variables anticipated to have a relationship with the magnitude of unobserved heterogeneity, including the dyad's initial trade and various country-specific factors: the structure of the economy (size of industry and services), the relative economic size (GDP sum and the difference in GDP), the countries connection to the global shipping network and its level of governance (average of the worldwide governance indicators). All variables were obtained from the World Bank's database. We used the information from the start of the study period or, in lack thereof, the first available data for each country pair. Due to the absence of exogeneity, these regressions do not aim to establish any causal relationship but only (conditional) correlations.

Table 4 lists the result using the estimates from the COMTRADE dataset (columns 1–4) and the ITPD-E dataset (columns 5–8). While these estimates come from the preferred specification, our findings are robust across different specifications. In Table 4, the odd columns show the results of the pooled regressions, while the even columns include importer and exporter fixed effects. As ϕ_{ij} is an estimated parameter, we also repeated the analysis using only significant values at the 5% level (columns 3, 4, 7 and 8). Other than increasing the coefficient of determination, the impact on the results was minimal.

We find that most explanatory variables of the gravity model are significantly correlated with the unobserved heterogeneity. Distance has a particularly strong effect, with countries that are further apart having a smaller response to θ_t . One notable exception is the FTA⁺ dummy, which is only significant when using the ϕ_{ij} s from the ITPD-E regressions. As we saw in Table 2, this seems to be specific to the COMTRADE dataset and the use of FTA⁺. As we already hinted at in the preceding section, domestic trade is associated with a strong negative effect. With a coefficient of minus 2, this implies a decrease that nullifies the positive coefficient on the domestic trade dummies. Common ethnographic languages are significantly positively correlated with ϕ_{ij} , although the effect is rather small.

Our investigation into the elements contributing to unobserved heterogeneity uncovers intriguing patterns. Particularly noteworthy is the positive and significant association with connectivity to the liner shipping network, which aligns with the idea that asymmetries in the evolution of trade costs constitute a pivotal aspect of unobserved heterogeneity. Second, the negative correlation with initial trade suggests that the magnitude of trade flows is converging between country pairs. Third, the economic structure of countries clearly also matters: larger country pairs of similar economic size and countries with a large share of industry tend to have a positive response to θ_t , while services-based economies see a negative impact. Finally, better governance has a conflicting but small impact, raising the unobserved heterogeneity for exporters and lowering it for importers.

²⁸ Keeping only those country pairs with at least 30 (ITPD-E) or 50 years (COMTRADE) reduces the sample size by two-thirds and removes four out of five dyads. However, the pattern in the θ_t s mostly retains its shape, and the correlation in their growth rate is around 50% for both datasets. Results are available upon request.

²⁹ The lowest correlation is when the distance parameter is excluded, which is 0.94%.

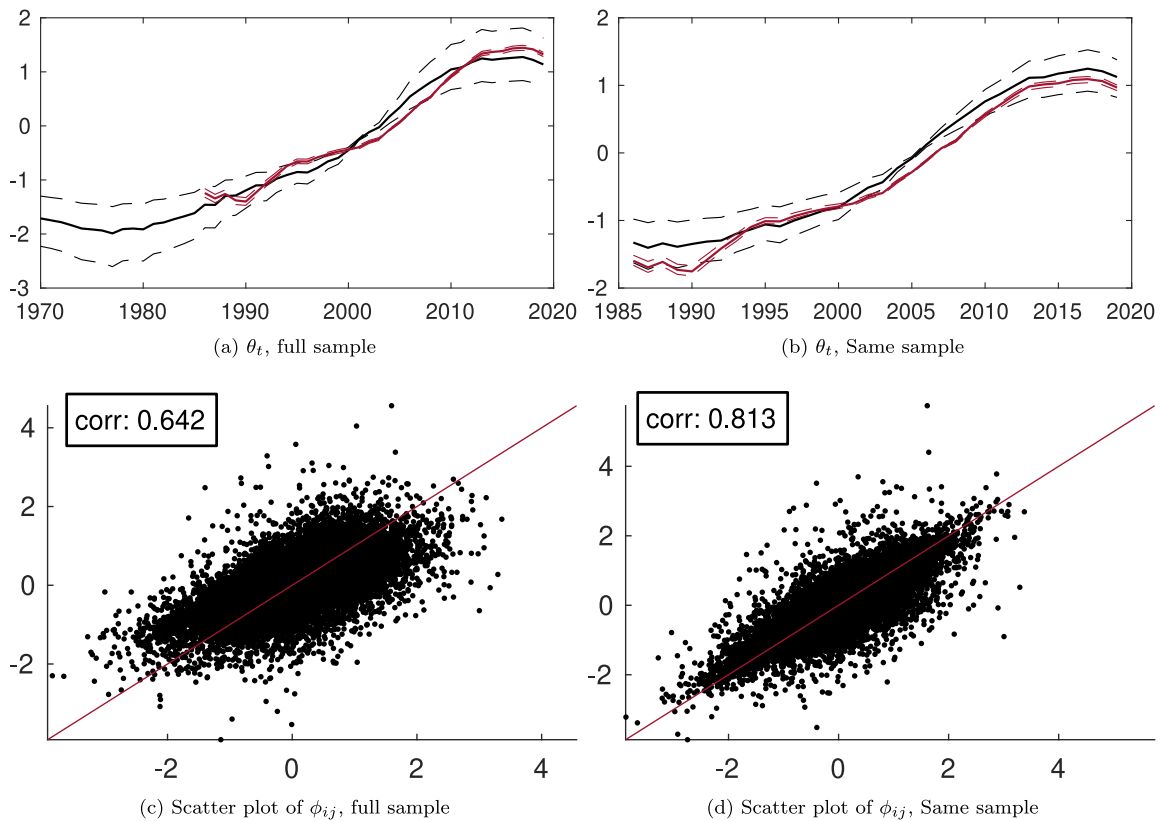


Fig. 6. Comparison of θ_t and ϕ_{ij} using COMTRADE and ITPD-E datasets.

NOTES: Parameter estimates of θ_t and ϕ_{ij} from the preferred specification. Panels a and b compare the values of θ_t between the COMTRADE (black) and ITPD-E datasets (red), together with their 95% confidence intervals (dotted lines). Panels c and d show the scatter plots of ϕ_{ij} with COMTRADE on the x-axis and ITPD-E on the y-axis. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

7. Explaining the growth in trade

The improvement in the model's precision also affects our ability to account for the growth in trade. For the ITPD-E dataset, the OLS model captures slightly more than half (55.1%) of the growth in trade over the last 30 years, although much of this is due to the inclusion of importer-year and exporter-year fixed effects. Once we include the correction for unobserved heterogeneity, this increases to 78%. As the COMTRADE dataset goes back further in time, we examine the growth in trade over the past 40 years. Here, the OLS model also captures around half (55.97%), but in our model this increases to 82.6%.

An increase in predictive power is perhaps not entirely surprising given the number of free parameters that our correction for unobserved heterogeneity adds to the model. To test whether the increase in the model goodness-of-fit outweighs the additional complexity, we ran a Bayes factor test. This is a typical test in Bayesian econometrics to see which of two competing models is most supported by the data. Crucially, it punishes models for overfitting the data in a way that, in large datasets, is similar to the Bayesian Information Criterion (Kass and Raftery, 1995). This punishment increases with the log of the sample size, meaning that the improvement in goodness-of-fit needs to be considerable for it to outweigh the increase in the number of parameters. Using the Bayes factor test, we compared our model with two specifications. The first was the log-linear model without unobserved heterogeneity. The second model was one in which θ_t followed a linear trend, similar to the model specified in Bergstrand et al. (2015). Both tests firmly rejected the hypotheses that the data preferred the alternatives: the log of the Bayes factor of the two tests was, respectively, 9.8×10^5 and 1.13×10^4 , vastly exceeding the traditional threshold of 2.³⁰ As such, we are confident in stating that the goodness of fit outweighs the increase in the model's complexity. Notwithstanding, when dealing with small datasets, the (log-)linear trend

³⁰ We also followed (Baier et al., 2014) and estimated a random growth model that assumes that the trends in the globalization process evolve linearly over time. The fixed effect in this model can be compared to our estimates ϕ_{ij} s. Estimating the random growth model using data at five-year intervals and comparing the fixed effects with the estimated ϕ_{ij} s that are significant at the 5% level, we find that the correlation between the bilateral time-varying heterogeneity in these two specifications is 0.56 for the COMTRADE data and 0.76 for the ITPD-E data. The higher correlation for the ITPD-E data is likely due to the shorter time horizon and a measure of θ that is more linear.

Table 4
Regressions on ϕ_{ij} .

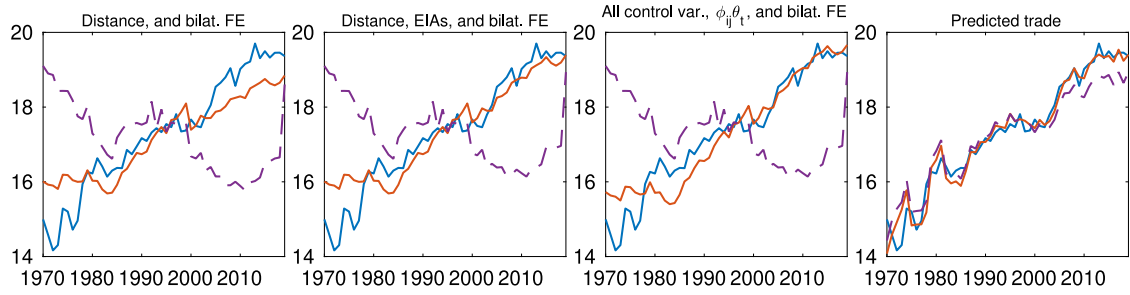
	COMTRADE				ITPD-E			
	All ϕ_{ij} s		Sig. ϕ_{ij} s		All ϕ_{ij} s		Sig. ϕ_{ij} s	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
lnDistw	−0.15 ^a (0.0073)	−0.20 ^a (0.0060)	−0.29 ^a (0.016)	−0.40 ^a (0.014)	−0.18 ^a (0.0070)	−0.22 ^a (0.0062)	−0.30 ^a (0.016)	−0.42 ^a (0.015)
Contiguity	−0.10 ^c (0.040)	−0.070 ^b (0.027)	−0.20 ^c (0.089)	−0.093 (0.062)	−0.21 ^a (0.038)	−0.12 ^a (0.027)	−0.29 ^b (0.090)	−0.17 ^c (0.067)
FTA ⁺	0.013 (0.016)	0.010 (0.013)	0.044 (0.034)	−0.017 (0.029)	−0.099 ^a (0.015)	−0.096 ^a (0.013)	−0.14 ^a (0.037)	−0.19 ^a (0.033)
Domestic	−0.28 ^b (0.086)	−0.61 ^a (0.068)	−0.88 ^a (0.21)	−1.94 ^a (0.17)	−0.69 ^a (0.081)	−1.01 ^a (0.068)	−1.20 ^a (0.18)	−2.20 ^a (0.16)
Language	0.041 ^b (0.013)	0.056 ^a (0.011)	0.060 ^c (0.027)	0.11 ^a (0.025)	0.034 ^b (0.012)	0.036 ^b (0.012)	0.059 ^c (0.029)	0.070 ^c (0.029)
Initial trade	−0.086 ^a (0.0018)	−0.087 ^a (0.0014)	−0.14 ^a (0.0036)	−0.14 ^a (0.0029)	−0.084 ^a (0.0017)	−0.093 ^a (0.0014)	−0.16 ^a (0.0039)	−0.17 ^a (0.0033)
GDPsum	0.045 ^a (0.0021)	–	0.084 ^a (0.0048)	–	0.045 ^a (0.0019)	–	0.086 ^a (0.0047)	–
GDPdiff	−0.018 ^a (0.0025)	–	−0.031 ^a (0.0054)	–	−0.016 ^a (0.0024)	–	−0.025 ^a (0.0056)	–
Industry _{ex}	0.054 (0.038)	–	0.073 (0.078)	–	0.21 ^a (0.037)	–	0.38 ^a (0.083)	–
Industry _{im}	−0.033 (0.038)	–	−0.071 (0.082)	–	0.13 ^a (0.037)	–	0.36 ^a (0.089)	–
Services _{ex}	−0.42 ^a (0.047)	–	−0.73 ^a (0.10)	–	−0.22 ^a (0.045)	–	−0.46 ^a (0.10)	–
Services _{im}	−0.20 ^a (0.047)	–	−0.38 ^a (0.10)	–	0.042 (0.045)	–	0.094 (0.11)	–
LinerShip _{ex}	0.54 ^a (0.026)	–	1.00 ^a (0.057)	–	0.33 ^a (0.025)	–	0.59 ^a (0.062)	–
LinerShip _{im}	0.47 ^a (0.026)	–	0.82 ^a (0.054)	–	0.27 ^a (0.025)	–	0.52 ^a (0.058)	–
WGI _{ex}	0.047 ^a (0.0067)	–	0.090 ^a (0.015)	–	0.030 ^a (0.0064)	–	0.078 ^a (0.015)	–
WGI _{im}	−0.0077 (0.0066)	–	0.00040 (0.014)	–	−0.017 ^b (0.0063)	–	−0.034 ^c (0.015)	–
Constant	0.47 ^a (0.11)	2.64 ^a (0.057)	0.70 ^b (0.24)	5.07 ^a (0.13)	−0.64 ^a (0.11)	1.76 ^a (0.054)	−1.50 ^a (0.26)	3.34 ^a (0.13)
Importer FE	–	yes	–	yes	–	yes	–	yes
Exporter FE	–	yes	–	yes	–	yes	–	yes
Obs.	19,161	32,594	8,058	12,307	18,790	31,335	6,857	10,825
R ²	14.2%	23.4%	22.8%	39.5%	14.5%	19.5%	25.2%	34.3%

NOTES: Regression of ϕ_{ij} on a number of control variables, where columns 1 through 4 use the COMTRADE data, and 5 through 8 the ITPD-E data. The odd-numbered columns are estimated using pooled OLS, while even-numbered columns include importer FE and exporter FE. In Columns 3, 4, 7, and 8, the sample is limited to those ϕ_{ij} s significantly different from zero at the 5% level. Standard errors between brackets. ^a, ^b, and ^c indicate significance at the 0.1%, 1%, and 5% level.

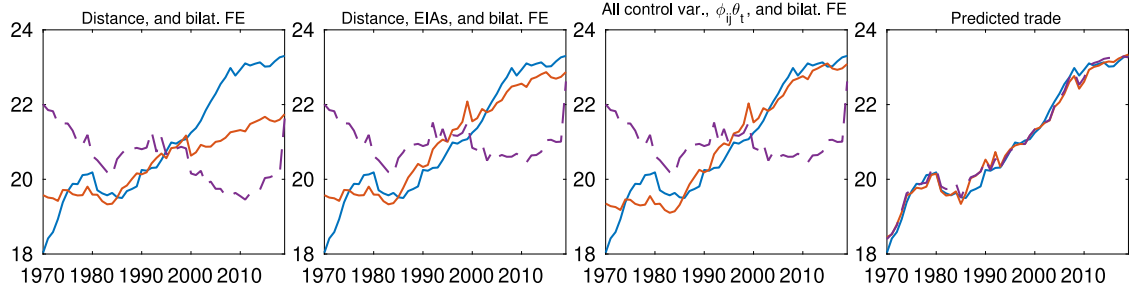
of the random growth model proposed [Bergstrand et al. \(2015\)](#) remains a suitable approximation. This is because the underlying unobserved heterogeneity can still be reasonably approximated by this simpler pattern, especially on shorter time scales.

To see what is driving this increase in explanatory power, [Fig. 7](#) plots out each variable's contribution to the model's goodness of fit for a number of country pairs. These predicted values are all based on the parameter estimates from our preferred specification. Each panel contrasts the measured trade flows (blue) with the predicted values from our model (red lines) and the standard OLS estimates (purple). The number of parameters used to predict trade increases from left to right. We start with distance and bilateral fixed effects in panel 1. Panel 2 adds the EIAs. Panel 3 adds all remaining time-invariant variables (contiguity, common language, and domestic trade dummies) as well as the unobserved heterogeneity. Panel 4 completes the model by adding in the importer-year and exporter-year fixed effects.

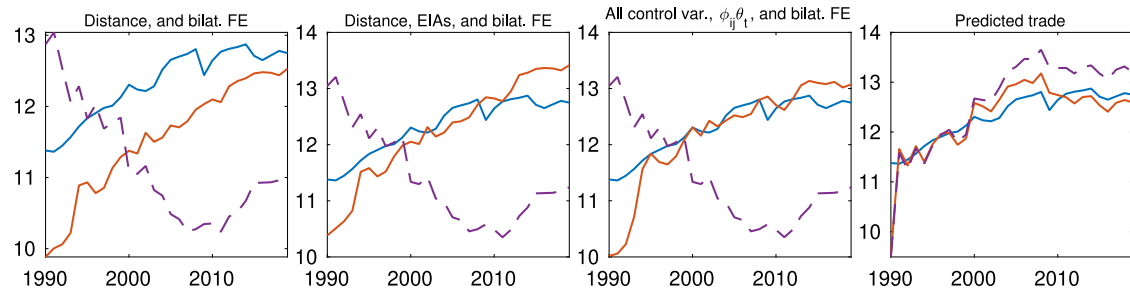
In [Fig. 7](#), we observe that both the COMTRADE and ITPD-E estimates demonstrate that the distance parameter accounts for a significant portion of the changes in trade flows within our regression results. The second most substantial contributors are the EIAs, resulting in predicted trade values that closely mirror the actual values. An illustrative example of this can be found in the second image in panel b, depicting Poland's exports to France steadily increasing from 2004 onward, coinciding with the former's accession to the EU. Conversely, the impact of other control variables remains relatively marginal. This holds true even for the unobserved heterogeneity component, which, for most countries, exerts only a minor influence (significantly differing from zero in only one of three cases). Therefore, while their inclusion is pivotal for the model, their actual contribution primarily lies in refining the parameter estimates of other variables.



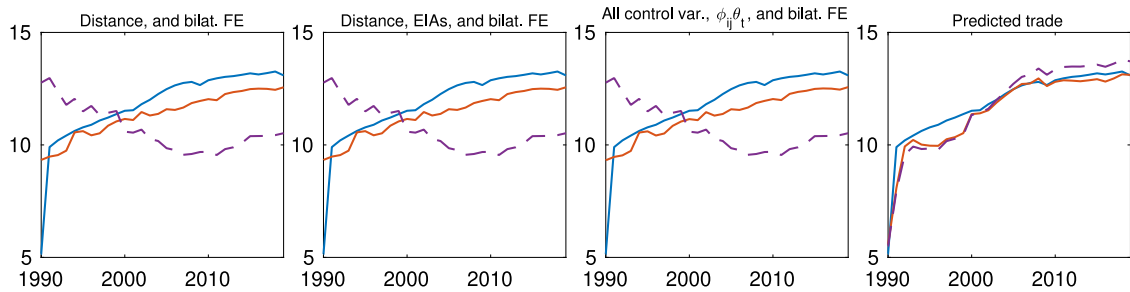
(a) Chile's exports to Austria (COMTRADE)



(b) Poland's exports to France (COMTRADE)



(c) Canada's exports to the USA (ITPD-E)



(d) China's exports to the USA (ITPD-E)

Fig. 7. Predicted versus measured trade growth.

NOTES: Measured (blue) versus the predicted trade with (red) and without (dashed purple) correcting for unobserved heterogeneity. The predicted trade flows are computed using the estimates from our preferred specification. Specifically, from left to right, the flows are predicted as follows (where $f_{ij}\theta_t$ is left out for the OLS estimates):

$$\hat{x}^1 = \text{dist}_{ij}\beta_t^d + \gamma_{ij};$$

$$\hat{x}^2 = \text{dist}_{ij}\beta_t^d + w_{ij,s}\alpha_r + \gamma_{ij};$$

$$\hat{x}^3 = z_{ij}\beta_t + w_{ij,s}\alpha_r + \gamma_{ij} + \phi_{ij}\theta_t + \delta_{i,t}^X;$$

$$\hat{x}^4 = z_{ij}\beta_t + w_{ij,s}\alpha_r + \gamma_{ij} + \phi_{ij}\theta_t + \delta_{i,t}^X + \delta_{j,t}^M. \text{ (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)}$$

This contrasts starkly with the OLS model estimates. Not only does the distance parameter exhibit its puzzling pattern, but its estimated impact also surpasses that of most other variables. Only when the importer-year and exporter-year dummies are reintroduced does the OLS prediction begin to resemble the actual trade values.

8. Conclusion

This paper has introduced a novel estimation strategy that leverages a gravity model with a Kalman filter to comprehensively analyze the dynamics of globalization and trade growth. By relaxing the traditional assumption about bilateral fixed effects and offering insights into the changing impacts of economic integration agreements on a year-by-year basis, this research has made significant contributions to the field. It also offers a novel way of addressing the “distance puzzle” by introducing greater heterogeneity in the response to globalization. Overall, this study offers a more nuanced understanding of international trade patterns and their evolution over time, paving the way for more accurate modeling of the forces shaping our global economic landscape.

While our investigation into the factors driving unobserved heterogeneity is still in an early stage, the regression results hint at several intriguing directions for future research. These avenues include delving into the economic size and structure of both the importing and exporting countries, as well as their connectivity within the shipping liner network. Other potential extensions involve broadening the model to encompass the PPML framework and utilizing historical trade data to assess how the unobserved global trend evolves over an extended period of time.

Data availability

[Baier_Standaert \(Data and code\)](#) (Mendeley Data)

Acknowledgments

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Appendix A. Robustness checks

A.1. Model specification

We experimented with over twenty different specifications of the gravity model, augmented with unobserved heterogeneity, on the two datasets (with and without restricting them to the same sample). The variations in specification primarily revolved around the inclusion or exclusion of certain variables, their interaction with time, and even the incorporation of additional variables, such as the sea and air distance from [Feyrer \(2019\)](#). We have omitted the details of these variations from the paper since they consistently yielded identical estimation results. The most substantial changes in the results were discussed in the paper: i.e., the introduction of the population-weighted distance parameter and the change from COMTRADE to ITPD-E data.

A.2. Spilling up the estimation sample

We reran the regression for the periods 1970–2000, 1980–2010 and 1990–2019. By way of illustration, [Fig. 8](#) plots the values of θ_t of the various subsets.³¹ As can be seen in [Table 5](#), the correlation between original θ_t and the values of the different sub-sample is in excess of 95%. However, since θ is non-stationary, the correlation between their levels is bound to be high. The correlation between the change in the θ_t s is lower but still positive and significant. The correlation between ϕ_{ij} s of the subset and baseline ϕ_{ij} s follows a similar pattern.

³¹ As the values of θ_t and ϕ_{ij} of the different models are normalized over different time-scales, a direct comparison between is not very illuminating. In order to make [Fig. 8](#), the values of θ_t were first set on an equal footing.

For example, to draw the values of θ_t for the 1970–2000 sub-sample, we computed the average and the variance of the estimates of the baseline model (θ_t) over the restricted sample period:

$$\bar{\theta} = \frac{1}{31} \sum_{t=1970}^{2000} \theta_t \quad s^2 = \frac{1}{30} \sum_{t=1970}^{2000} (\theta_t - \bar{\theta})^2$$

The normalized values of θ_t^{70} can then be computed as: $\bar{\theta}_t^{70} = \theta_t^{70} * s^2 + \bar{\theta}$.

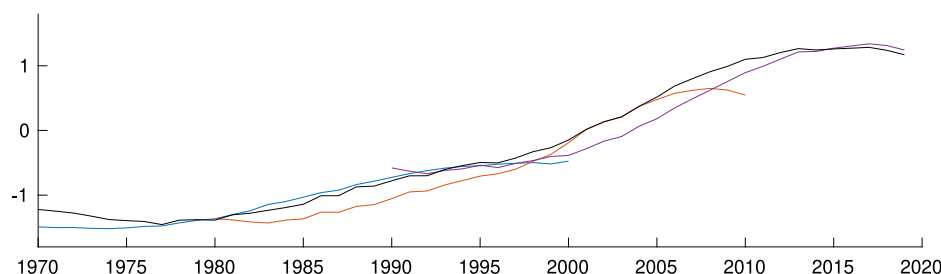


Fig. 8. Estimates of θ_i for different periods.

Notes: Plot of θ_i of the baseline model (black line), and the 1970–2000 (blue line), 1980–2010 (red line), and 1990–2019 subsets (purple line). Using the specification from column 3 of Table 2 (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5

Correlation between the baseline model and robustness checks.

	θ_i (1)	$\Delta\theta_i$ (2)	ϕ_{ij} (3)
a. Different subsets			
1970–2000	95.3%	55.9%	46.2%
1980–2010	98.6%	58.0%	78.8%
1990–2019	98.0%	75.6%	74.8%
b. Interval data			
3-yearly	99.9%	98.2%	88.2%
5-yearly	99.9%	94.1%	82.8%

Correlation between the results of the full sample and robustness checks using the specification from column 3 of Table 2. Column 1 compares the values of θ_i ; column 2 compares the change in θ_i on a 3-yearly (row 1), 5-yearly (row 2) or yearly basis (row 3–5); and column 3 compares the values of ϕ_{ij} .

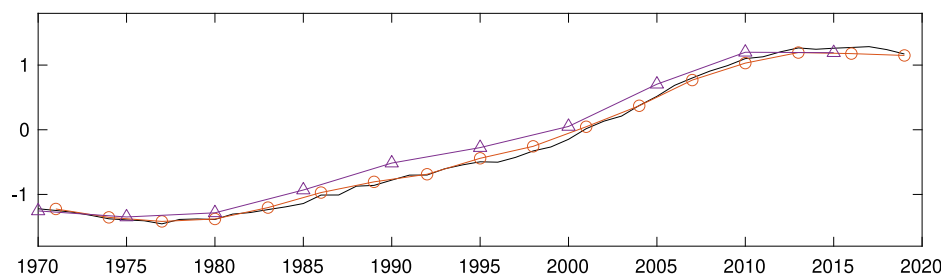


Fig. 9. Plot of θ_i s using yearly and interval data.

Notes: Plot of θ_i of the baseline model (black line), 3-yearly (red circles), and 5-yearly model (purple triangles). Using the specification from column 3 of Table 2.

A.3. Using interval data

We also ran the model using one in every 3 or 5 observations. As can be seen from panel b of Table 5, this had little effect on the estimates of θ_i . Even the changes in θ_i are barely affected. The correlation between the ϕ_{ij} s of the different models is slightly lower, most likely due to the reduction in number of observations that can be used to estimate the bilateral trade cost variables (see Fig. 9).

A.4. Non-hierarchical priors

The choice for hierarchical priors on the bilateral trade cost variables, γ_{ij} and ϕ_{ij} , did not fundamentally alter our findings. As can be seen in panel a of Fig. 10, the parameter estimates of the bilateral fixed costs are not meaningfully affected by the choice of prior. The values of ϕ_{ij} are more affected but remain highly correlated. As panel b shows, all scatter points lie between the x -axis and the 45° line, which means that the use of hierarchical priors shrinks the estimates down. As we would expect, the biggest changes are associated with country pairs with the fewest observations.

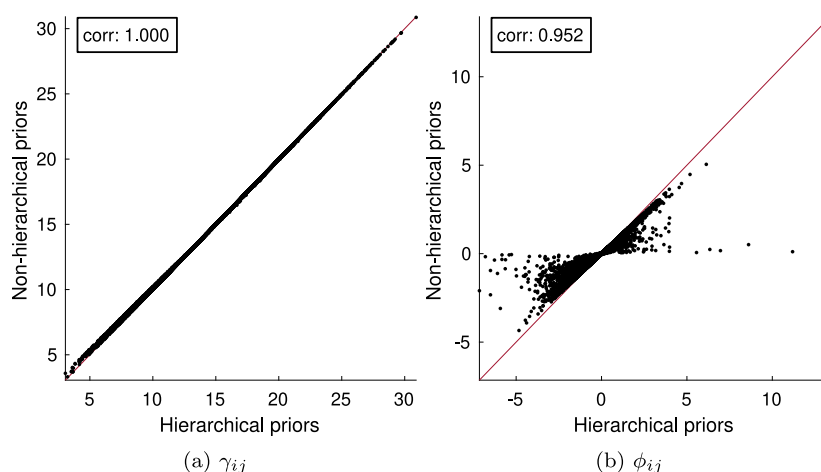


Fig. 10. Comparison of ϕ_{ij} and γ_{ij} with and without hierarchical priors.

Notes: Scatter plot of the values of the trade cost variables using non-hierarchical priors (y-axis) and hierarchical priors (x-axis). The red line shows the 45° line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Appendix B. Estimation algorithm

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.euroecorev.2024.104679>.

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